Government College of Engineering, Jalgaon
(An Autonomous Institute of Government of Maharashtra)

Name :	**PRN :**
Class : L. Y. B.Tech Computer	**Batch:**
Subject : CO456U Cloud Computing Lab	**Sem :** VIIIth
Course Teacher: Prof. S. D. Cheke	**Academic Year :** 2025-26
Date of Performance :	**Date of Completion :**

Practical No. 3

Aim: Create and deploy news feed application on aws/gcp. Please also show how we can install database on instance and connect it to application.

Requirements-

- Cloud Account (Amazon Web Services / Google Cloud Platform)
- Virtual Machine (EC2 / Compute Engine)
- OS: Ubuntu Linux
- Python installed
- Flask framework
- Database: MySQL / PostgreSQL
- SSH client (e.g., MobaXterm / Terminal)
- Internet connection
- Basic knowledge of Python & SQL

Theory:

What is a News Feed Application?

A News Feed Application is a web app that:

- Displays latest posts/news
- Stores data in a database
- Retrieves and shows data dynamically

👉 Example:

- Users add news → stored in DB → displayed on website

PRACTICAL STEPS

STEP 1: Create AWS Account

1. Go to: Amazon Web Services
2. Sign up using email
3. Add debit/credit card
4. Login to AWS Console

STEP 2: Launch EC2 Instance (Virtual Server)

1. In AWS Console → Search EC2
2. Open Amazon EC2
3. Click Launch Instance

Fill details:

- Name → NewsAppServer
- AMI (OS) → Ubuntu 22.04
- Instance Type → t2.micro (Free tier)

Create Key Pair

- Click Create Key Pair
- Name → news-key
- Type → .pem
- Download file (IMPORTANT ⚠️)

Network Settings

Click Edit → Add rules:

Type	Port
SSH	22
HTTP	80
Custom	5000

Click Launch Instance

STEP 3: Connect to EC2

Using Terminal / MobaXterm

Go to folder where .pem file is:

chmod 400 news-key.pem

Connect:

ssh -i news-key.pem ubuntu@YOUR_PUBLIC_IP

After login → you are inside Linux server

STEP 4: Update Server
sudo apt update
sudo apt upgrade -y

STEP 5: Install Python & Tools
sudo apt install python3 python3-pip git -y
Check version:
python3 --version

STEP 6: Install Flask & MySQL Connector
pip3 install flask mysql-connector-python

STEP 7: Install MySQL Database (IMPORTANT)
sudo apt install mysql-server -y
Start MySQL:
sudo systemctl start mysql
sudo systemctl enable mysql

STEP 8: Secure MySQL
sudo mysql_secure_installation
👉 Answer like this:

- Set root password → YES
- Remove anonymous users → YES
- Disallow root login remotely → YES
- Remove test DB → YES

STEP 9: Create Database & Table
Login:
sudo mysql
Now run:

```
CREATE DATABASE newsdb;

USE newsdb;

CREATE TABLE news (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255),
    content TEXT
);
```

👉 Create Database User

```
CREATE USER 'newsuser'@'localhost' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON newsdb.* TO 'newsuser'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

STEP 10: Create Application Code

Create file:

nano app.py

Paste:

```
from flask import Flask, request, jsonify
import mysql.connector

app = Flask(__name__)

db = mysql.connector.connect(
    host="localhost",
    user="newsuser",
    password="password",
    database="newsdb"
)

@app.route('/')
def home():
    return "News Feed App Running Successfully!"

@app.route('/add', methods=['POST'])
def add_news():
    data = request.json
    cursor = db.cursor()
    cursor.execute("INSERT INTO news (title, content) VALUES (%s, %s)",
                   (data['title'], data['content']))
    db.commit()
    return jsonify({"message": "News added"})

@app.route('/news', methods=['GET'])
def get_news():
    cursor = db.cursor()
    cursor.execute("SELECT * FROM news")
    data = cursor.fetchall()
    return jsonify(data)

app.run(host='0.0.0.0', port=5000)
```

Save:

CTRL + X → Y → ENTER

STEP 11: Run Application

python3 app.py

👉 Output:

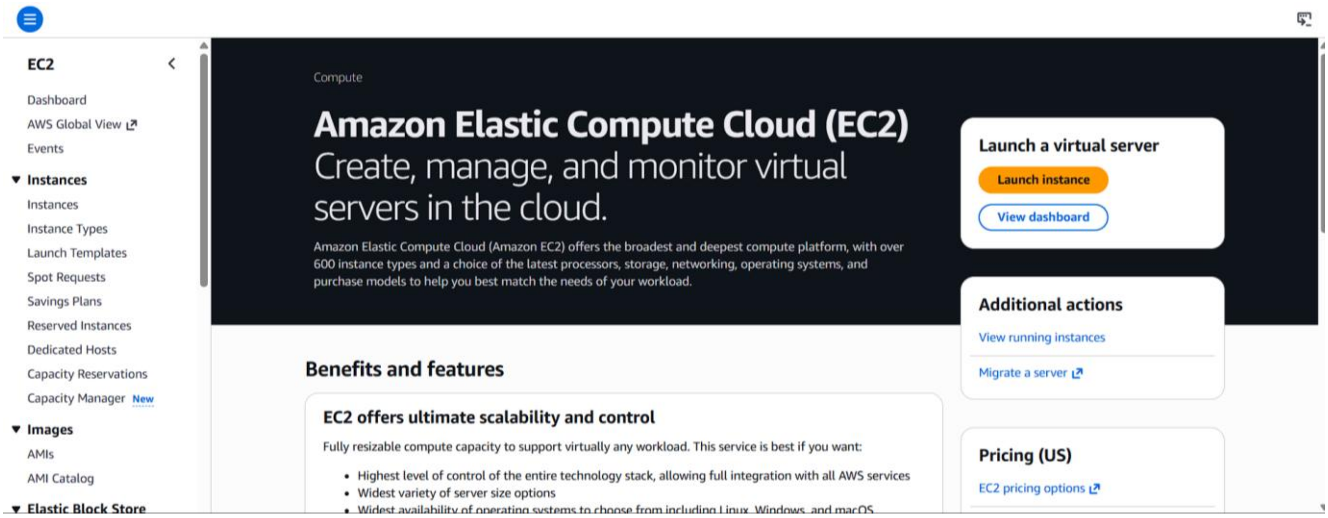
Running on http://0.0.0.0:5000

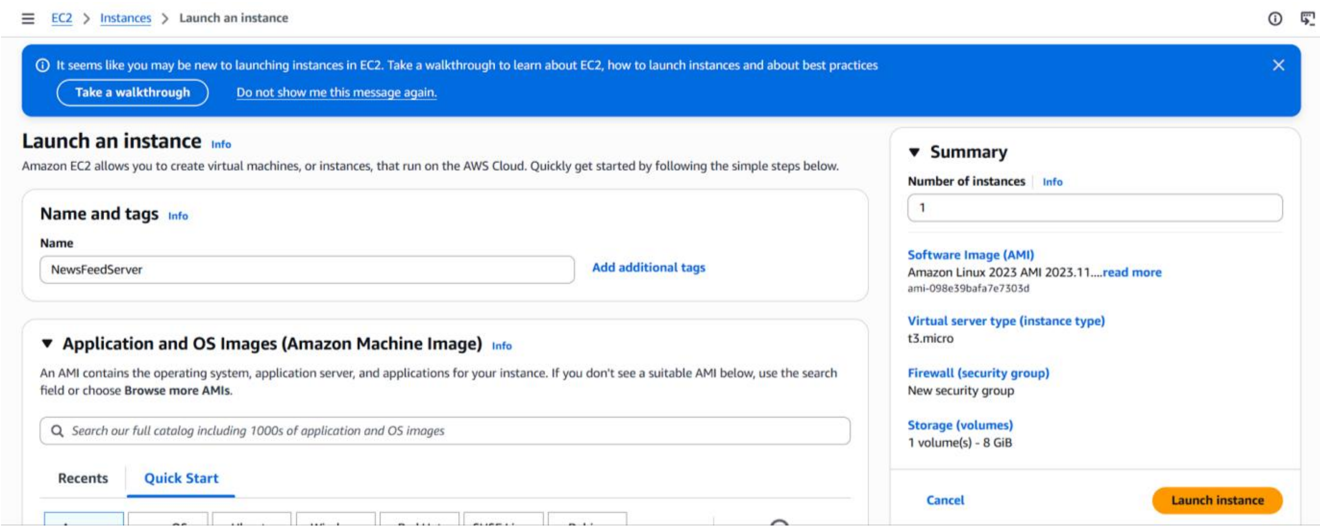
STEP 12: Access in Browser
Open:
http://YOUR_PUBLIC_IP:5000
👉 You will see:
News Feed App Running Successfully!

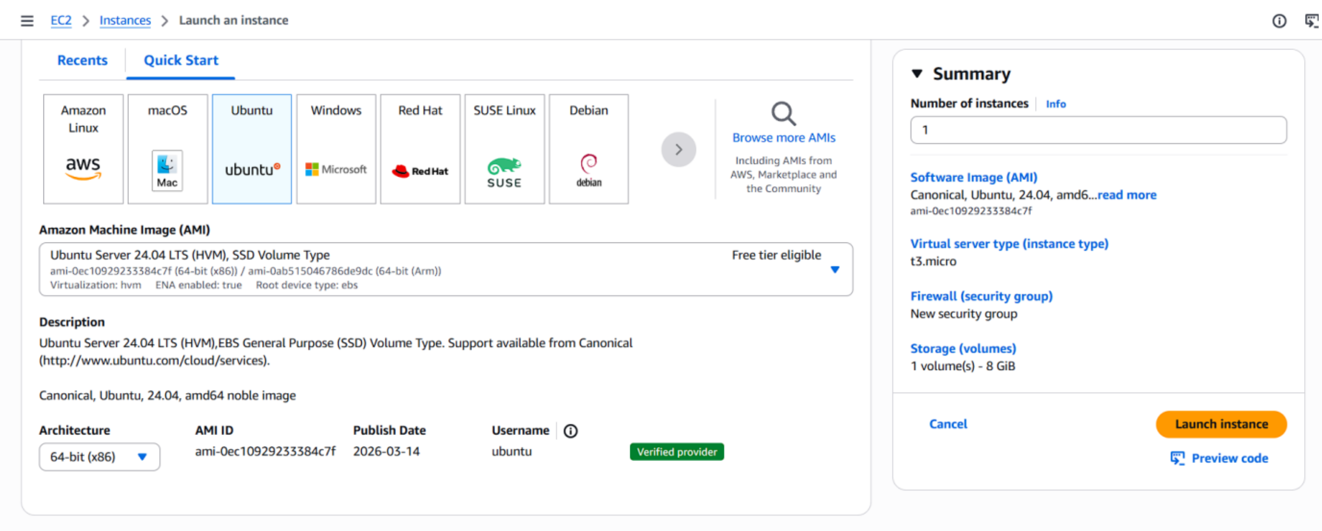
STEP 13: Test API (IMPORTANT)
👉 Add News
Use Postman:
POST http://YOUR_IP:5000/add
Body:
```
{
  "title": "First News",
  "content": "This is my cloud app"
}
```

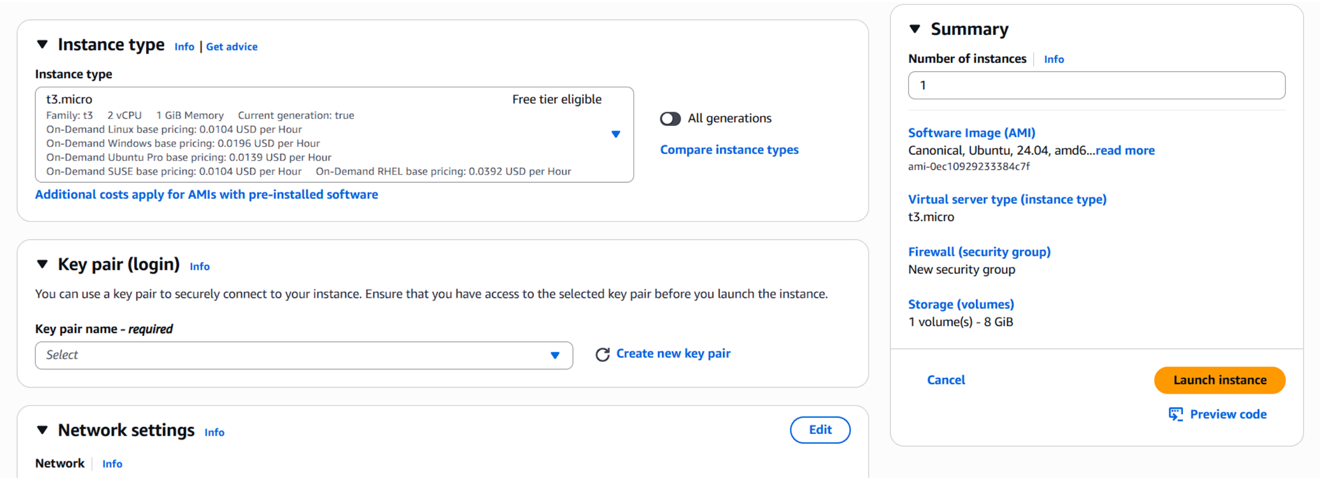
👉 Get News
GET http://YOUR_IP:5000/news

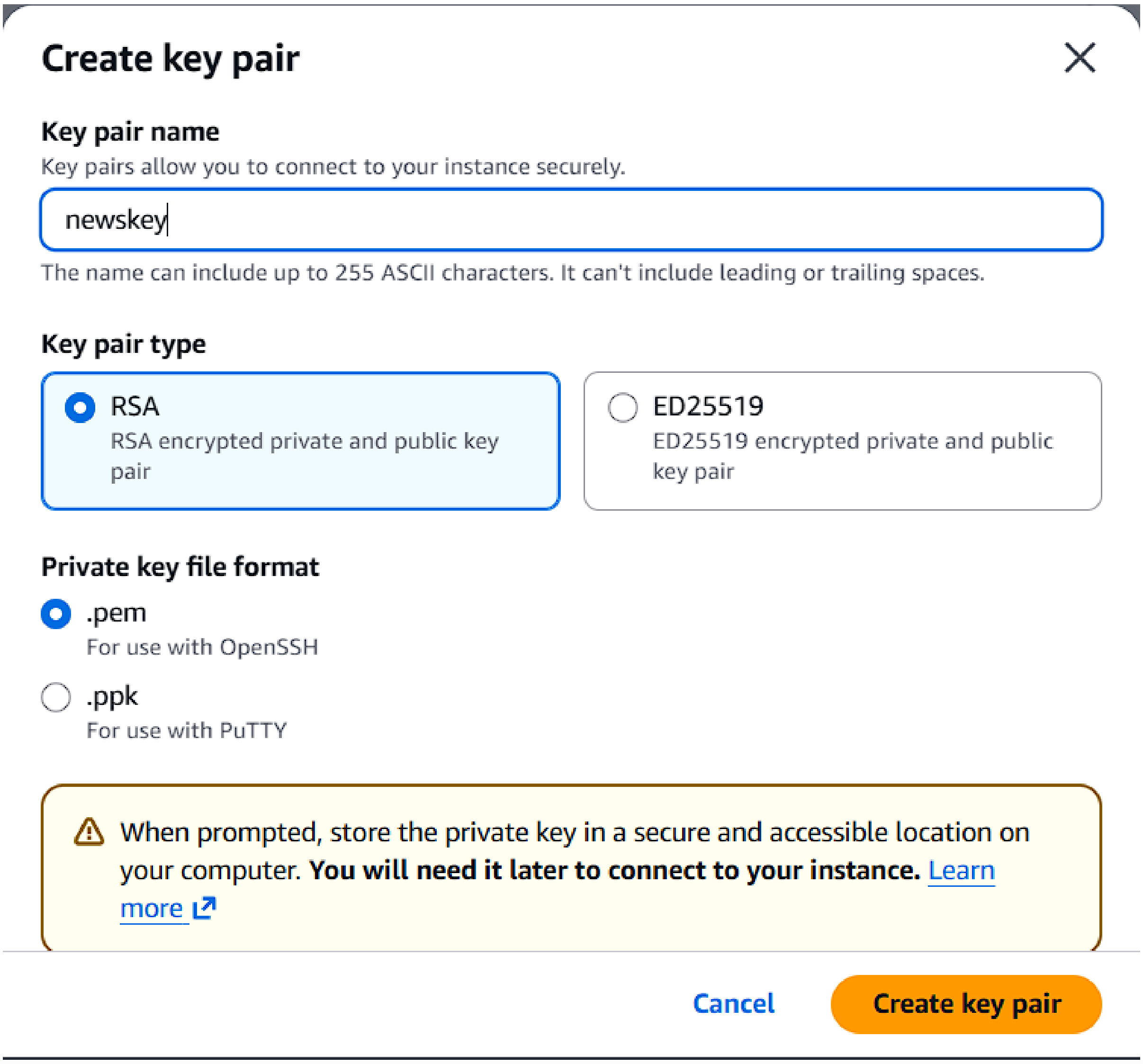
Output:-



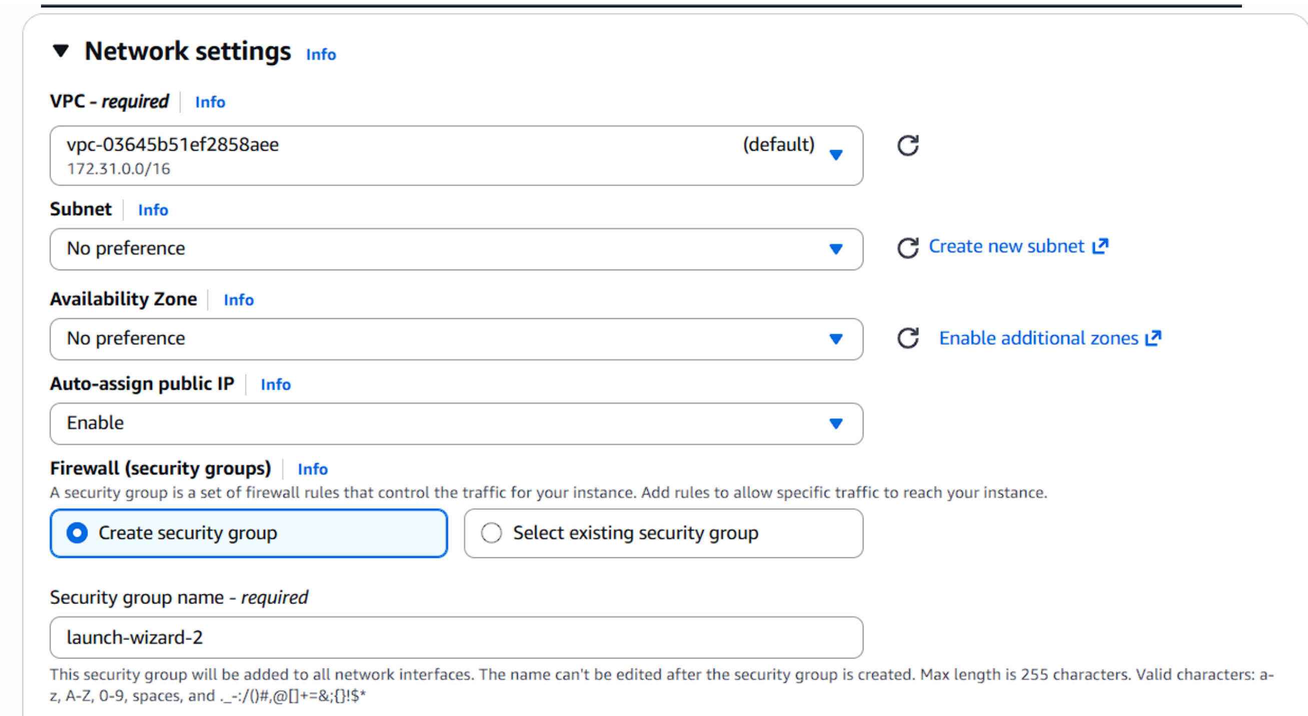

EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
NewsFeedServer
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-098e39bafa7e7303d
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance


EC2 > Instances > Launch an instance
Recents
Quick Start
Amazon Linux
aws
macOS
Mac
Ubuntu
ubuntu
Windows
Microsoft
Red Hat
Red Hat
SUSE Linux
SUSE
Debian
debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
Free tier eligible
ami-0ec10929233384c7f (64-bit (x86)) / ami-0ab515046786de9dc (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Canonical, Ubuntu, 24.04, amd64 noble image
Architecture
64-bit (x86)
AMI ID
ami-0ec10929233384c7f
Publish Date
2026-03-14
Username
ubuntu
Verified provider
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0ec10929233384c7f
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

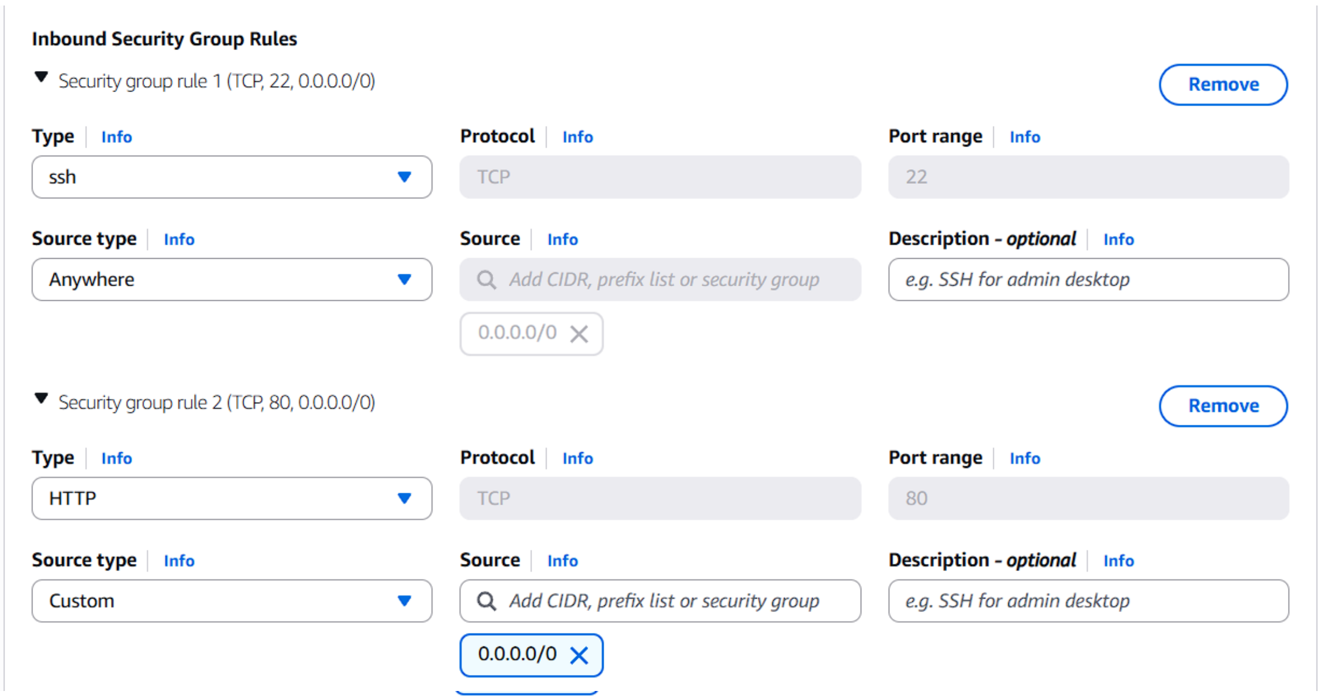

Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Select
Create new key pair
Network settings Info
Edit
Network Info
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0ec10929233384c7f
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code



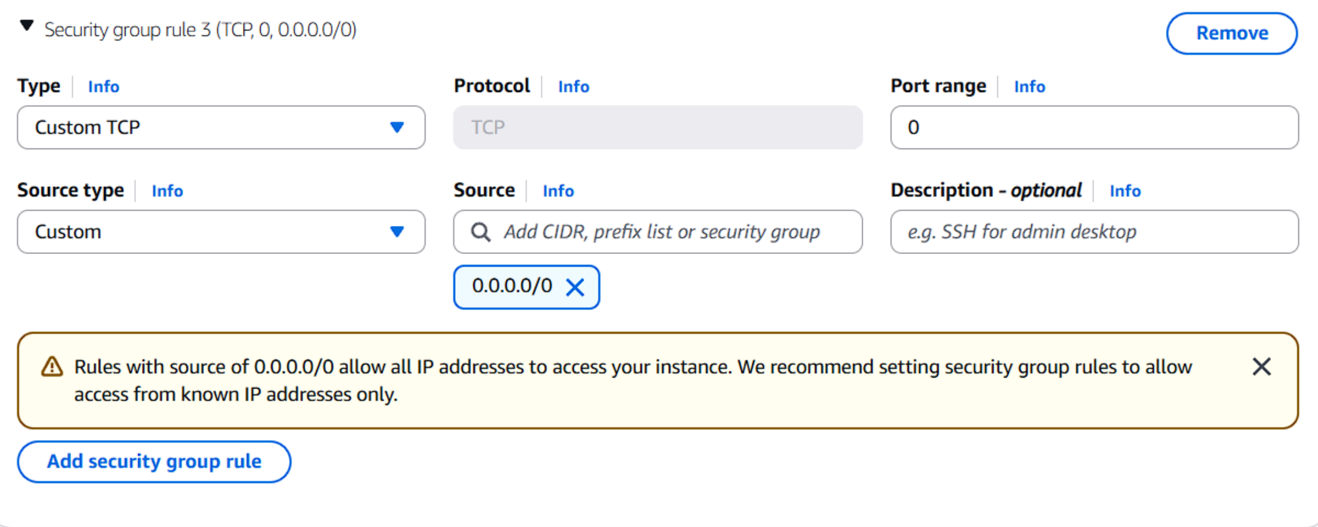
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
newskey
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair



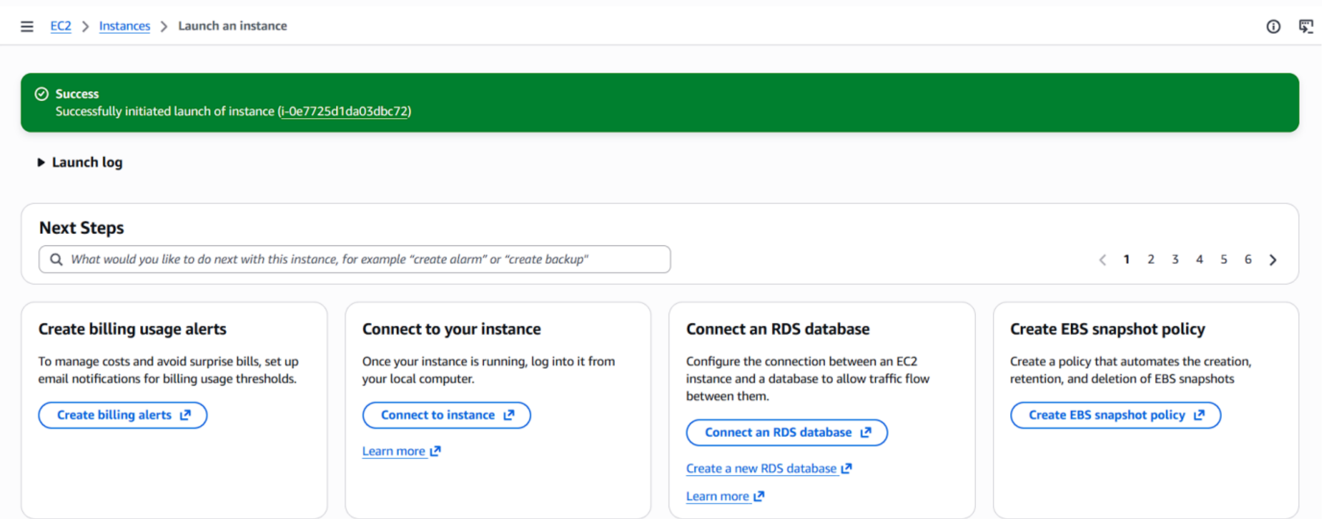
Network settings
Info
VPC - required
Info
vpc-03645b51ef2858aee
172.31.0.0/16
(default)
Subnet
Info
No preference
Create new subnet
Availability Zone
Info
No preference
Enable additional zones
Auto-assign public IP
Info
Enable
Firewall (security groups)
Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
launch-wizard-2
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*



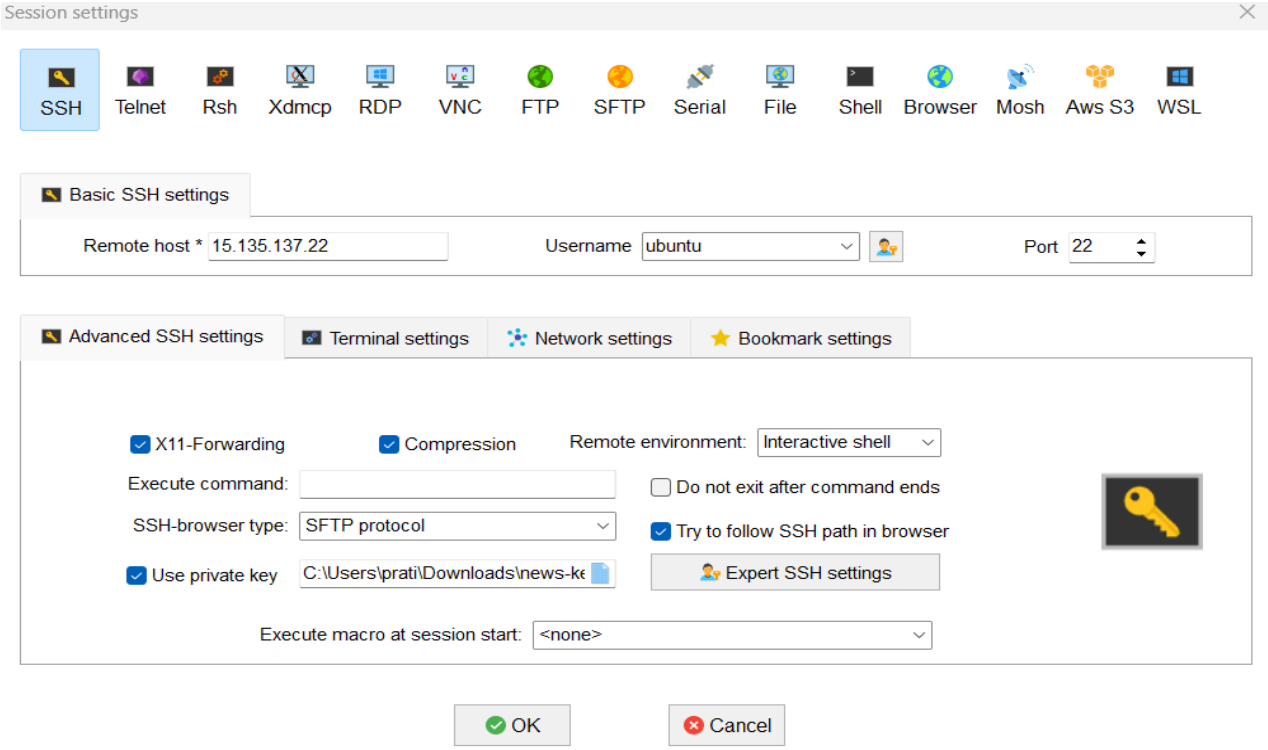
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 0.0.0.0/0)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Security group rule 2 (TCP, 80, 0.0.0.0/0)
Remove
Type Info
HTTP
Protocol Info
TCP
Port range Info
80
Source type Info
Custom
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop

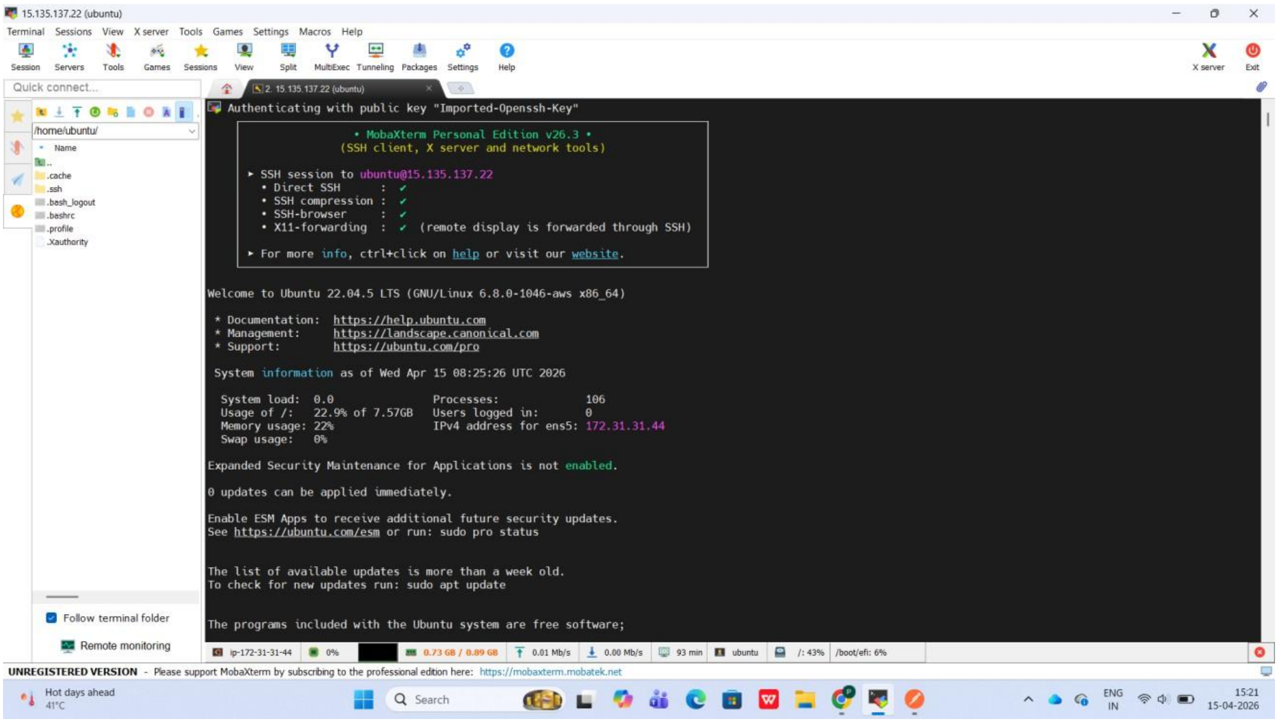


Security group rule 3 (TCP, 0, 0.0.0.0/0)
Remove
Type Info
Custom TCP
Protocol Info
TCP
Port range Info
0
Source type Info
Custom
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
e.g. SSH for admin desktop
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Add security group rule



EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-0e7725d1da03dbc72)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy


Session settings
SSH
Telnet
Rsh
Xdmcp
RDP
VNC
FTP
SFTP
Serial
File
Shell
Browser
Mosh
Aws S3
WSL
Basic SSH settings
Remote host * 15.135.137.22
Username ubuntu
Port 22
Advanced SSH settings
Terminal settings
Network settings
Bookmark settings
X11-Forwarding
Compression
Remote environment: Interactive shell
Execute command:
Do not exit after command ends
SSH-browser type: SFTP protocol
Try to follow SSH path in browser
Use private key
C:\Users\prati\Downloads\news-ke
Expert SSH settings
Execute macro at session start: <none>
OK
Cancel


15.135.137.22 (ubuntu)
Authenticating with public key "Imported-Openssh-Key"
• MobaXterm Personal Edition v26.3 •
(SSH client, X server and network tools)
▸ SSH session to ubuntu@15.135.137.22
• Direct SSH : ✔
• SSH compression : ✔
• SSH-browser : ✔
• X11-forwarding : ✔ (remote display is forwarded through SSH)
▸ For more info, ctrl+click on help or visit our website.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1046-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Wed Apr 15 08:25:26 UTC 2026
System load: 0.0 Processes: 106
Usage of /: 22.9% of 7.57GB Users logged in: 0
Memory usage: 22% IPv4 address for ens5: 172.31.31.44
Swap usage: 0%
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
Follow terminal folder
Remote monitoring
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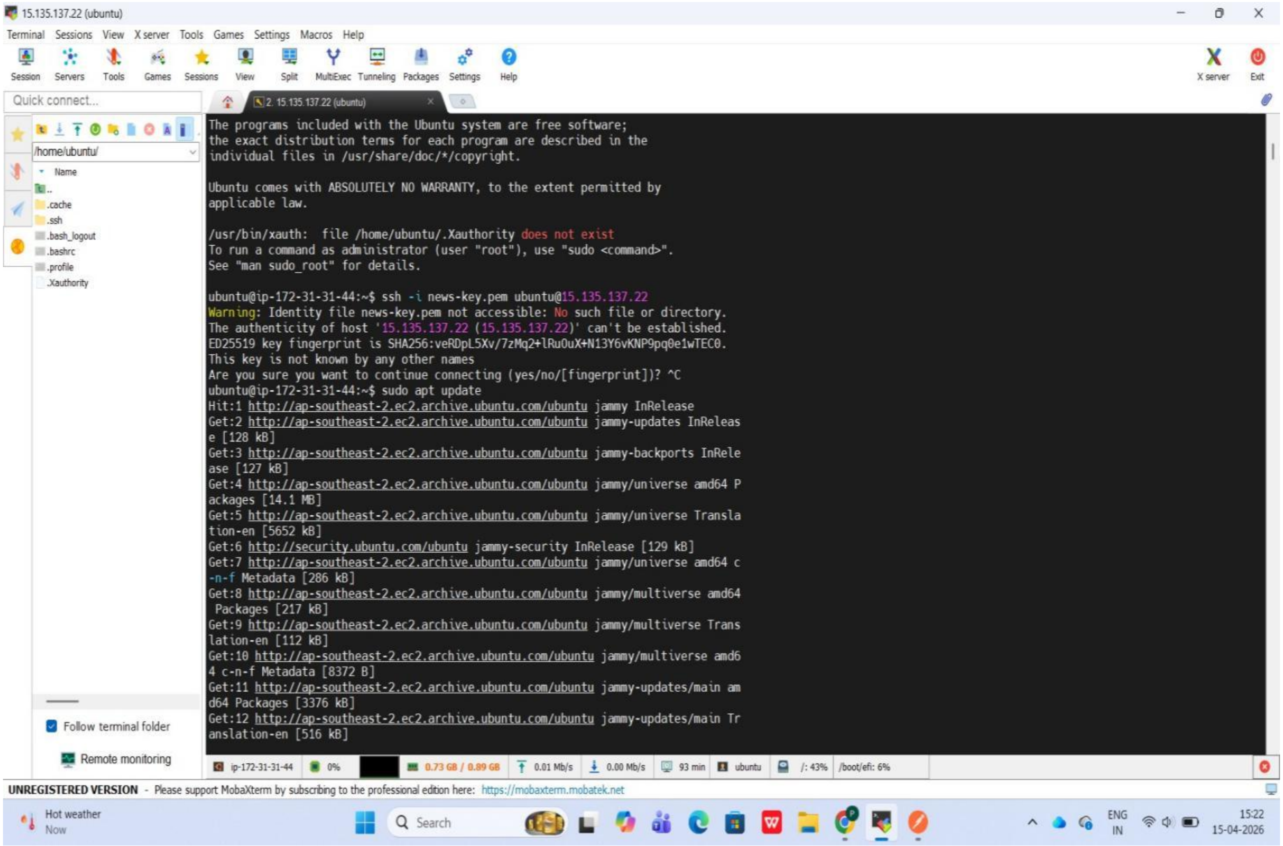
15.135.137.22 (ubuntu)
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
/usr/bin/xauth: file /home/ubuntu/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-172-31-31-44:~$ ssh -i news-key.pem ubuntu@15.135.137.22
Warning: Identity file news-key.pem not accessible: No such file or directory.
The authenticity of host '15.135.137.22 (15.135.137.22)' can't be established.
ED25519 key fingerprint is SHA256:veRDpL5Xv/7zMq2+lRu0uX+N13Y6vKNP9pq0e1wTEC0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? ^C
ubuntu@ip-172-31-31-44:~$ sudo apt update
Hit:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:5 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:7 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3376 kB]
Get:12 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [516 kB]
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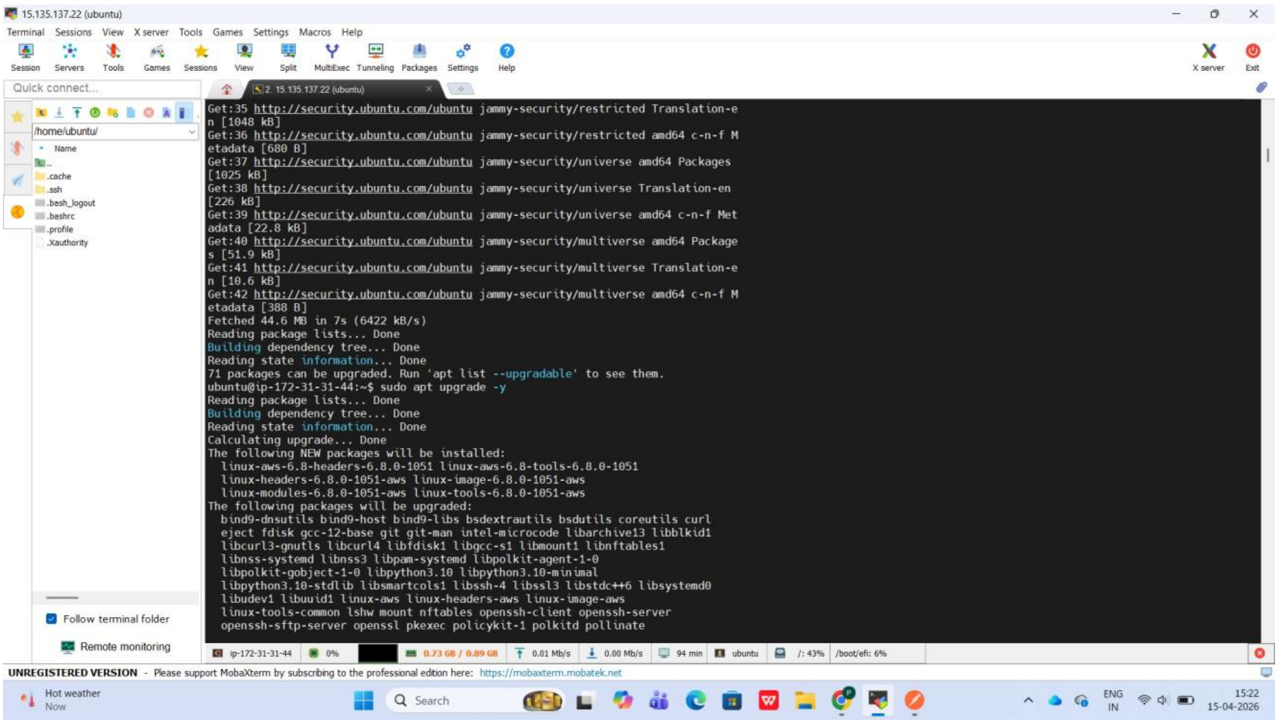
15.135.137.22 (ubuntu)
Get:35 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [1048 kB]
Get:36 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 c-n-f Metadata [680 B]
Get:37 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1025 kB]
Get:38 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [226 kB]
Get:39 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:40 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [51.9 kB]
Get:41 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [10.6 kB]
Get:42 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [388 B]
Fetched 44.6 MB in 7s (6422 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-31-44:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
linux-aws-6.8-headers-6.8.0-1051 linux-aws-6.8-tools-6.8.0-1051
linux-headers-6.8.0-1051-aws linux-image-6.8.0-1051-aws
linux-modules-6.8.0-1051-aws linux-tools-6.8.0-1051-aws
The following packages will be upgraded:
bind9-dnsutils bind9-host bind9-libs bsdextrautils bsdutils coreutils curl
eject fdisk gcc-12-base git git-man intel-microcode libarchive13 libblkid1
libcurl3-gnutls libcurl4 libfdisk1 libgcc-s1 libmount1 libnftables1
libnss-systemd libnss3 libpam-systemd libpolkit-agent-1-0
libpolkit-gobject-1-0 libpython3.10 libpython3.10-minimal
libpython3.10-stdlib libsmartcols1 libssh-4 libssl3 libstdc++6 libsystemd0
libudev1 libuuid1 linux-aws linux-headers-aws linux-image-aws
linux-tools-common lshw mount nftables openssh-client openssh-server
openssh-sftp-server openssl pkexec policykit-1 polkitd pollinate
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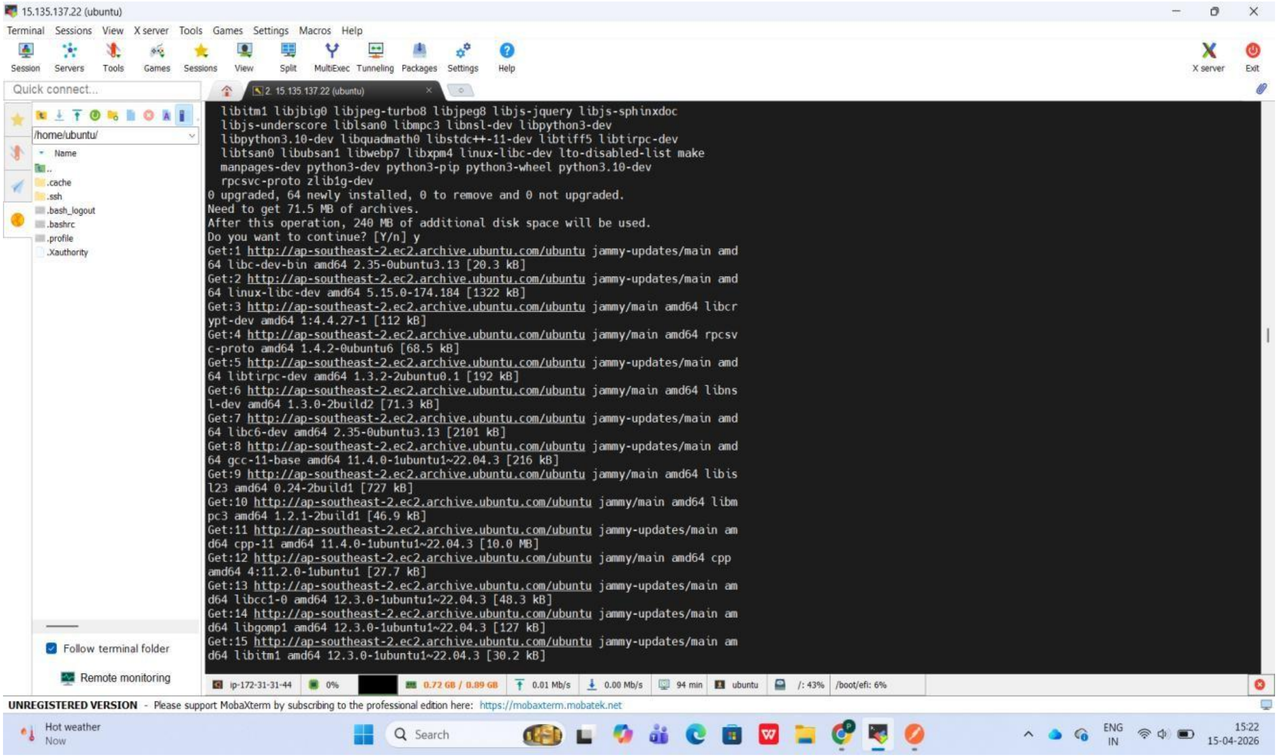
15.135.137.22 (ubuntu)
libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc
libjs-underscore liblsan0 libmpc3 libnsl-dev libpython3-dev
libpython3.10-dev libquadmath0 libstdc++-11-dev libtiff5 libtirpc-dev
libtsan0 libubsan1 libwebp7 libxpm4 linux-libc-dev lto-disabled-list make
manpages-dev python3-dev python3-pip python3-wheel python3.10-dev
rpcsvc-proto zlib1g-dev
0 upgraded, 64 newly installed, 0 to remove and 0 not upgraded.
Need to get 71.5 MB of archives.
After this operation, 240 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-dev-bin amd64 2.35-0ubuntu3.13 [20.3 kB]
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-libc-dev amd64 5.15.0-174.184 [1322 kB]
Get:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libcrypt-dev amd64 1:4.4.27-1 [112 kB]
Get:4 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 rpcsvc-proto amd64 1.4.2-0ubuntu6 [68.5 kB]
Get:5 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtirpc-dev amd64 1.3.2-2ubuntu0.1 [192 kB]
Get:6 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libnsl-dev amd64 1.3.0-2build2 [71.3 kB]
Get:7 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-dev amd64 2.35-0ubuntu3.13 [2101 kB]
Get:8 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11-base amd64 11.4.0-1ubuntu1~22.04.3 [216 kB]
Get:9 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libisl23 amd64 0.24-2build1 [727 kB]
Get:10 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libmpc3 amd64 1.2.1-2build1 [46.9 kB]
Get:11 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 cpp-11 amd64 11.4.0-1ubuntu1~22.04.3 [10.0 MB]
Get:12 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 cpp amd64 4:11.2.0-1ubuntu1 [27.7 kB]
Get:13 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcc1-0 amd64 12.3.0-1ubuntu1~22.04.3 [48.3 kB]
Get:14 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgomp1 amd64 12.3.0-1ubuntu1~22.04.3 [127 kB]
Get:15 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libitm1 amd64 12.3.0-1ubuntu1~22.04.3 [30.2 kB]
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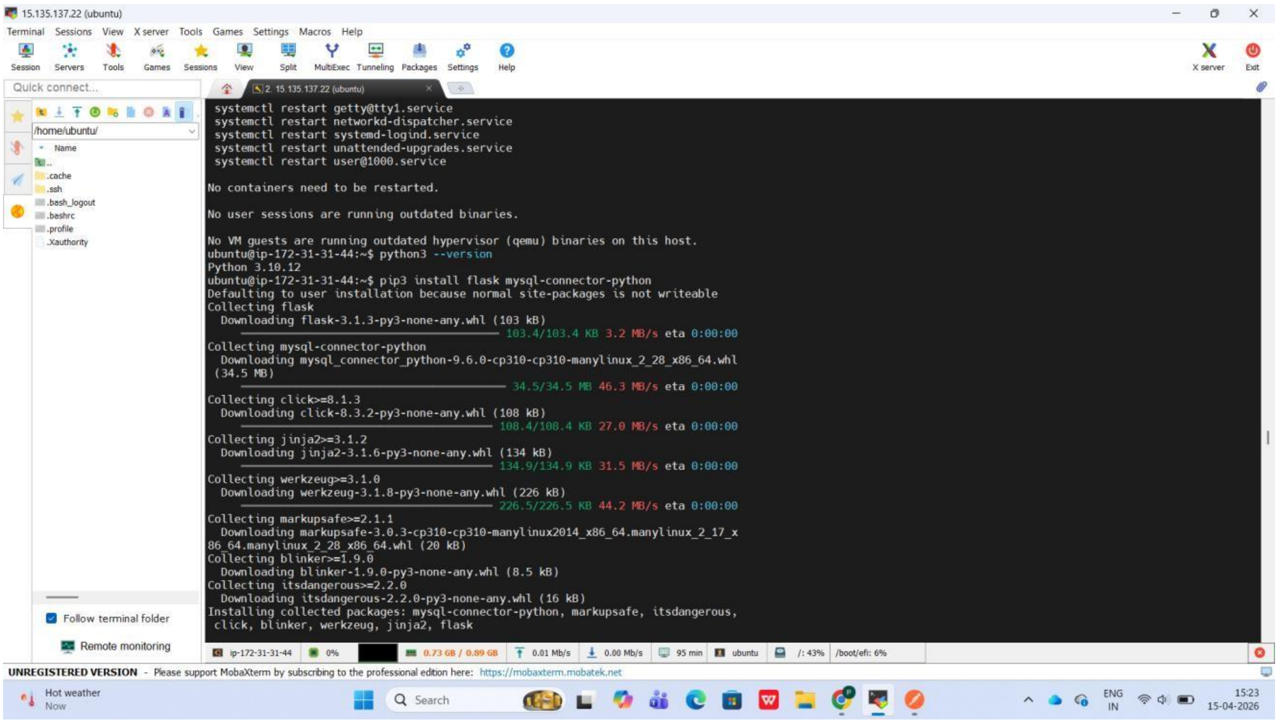
15.135.137.22 (ubuntu)
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service
systemctl restart user@1000.service
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-31-44:~$ python3 --version
Python 3.10.12
ubuntu@ip-172-31-31-44:~$ pip3 install flask mysql-connector-python
Defaulting to user installation because normal site-packages is not writeable
Collecting flask
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
103.4/103.4 KB 3.2 MB/s eta 0:00:00
Collecting mysql-connector-python
Downloading mysql_connector_python-9.6.0-cp310-cp310-manylinux_2_28_x86_64.whl (34.5 MB)
34.5/34.5 MB 46.3 MB/s eta 0:00:00
Collecting click>=8.1.3
Downloading click-8.3.2-py3-none-any.whl (108 kB)
108.4/108.4 KB 27.0 MB/s eta 0:00:00
Collecting jinja2>=3.1.2
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 KB 31.5 MB/s eta 0:00:00
Collecting werkzeug>=3.1.0
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
226.5/226.5 KB 44.2 MB/s eta 0:00:00
Collecting markupsafe>=2.1.1
Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting blinker>=1.9.0
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting itsdangerous>=2.2.0
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Installing collected packages: mysql-connector-python, markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
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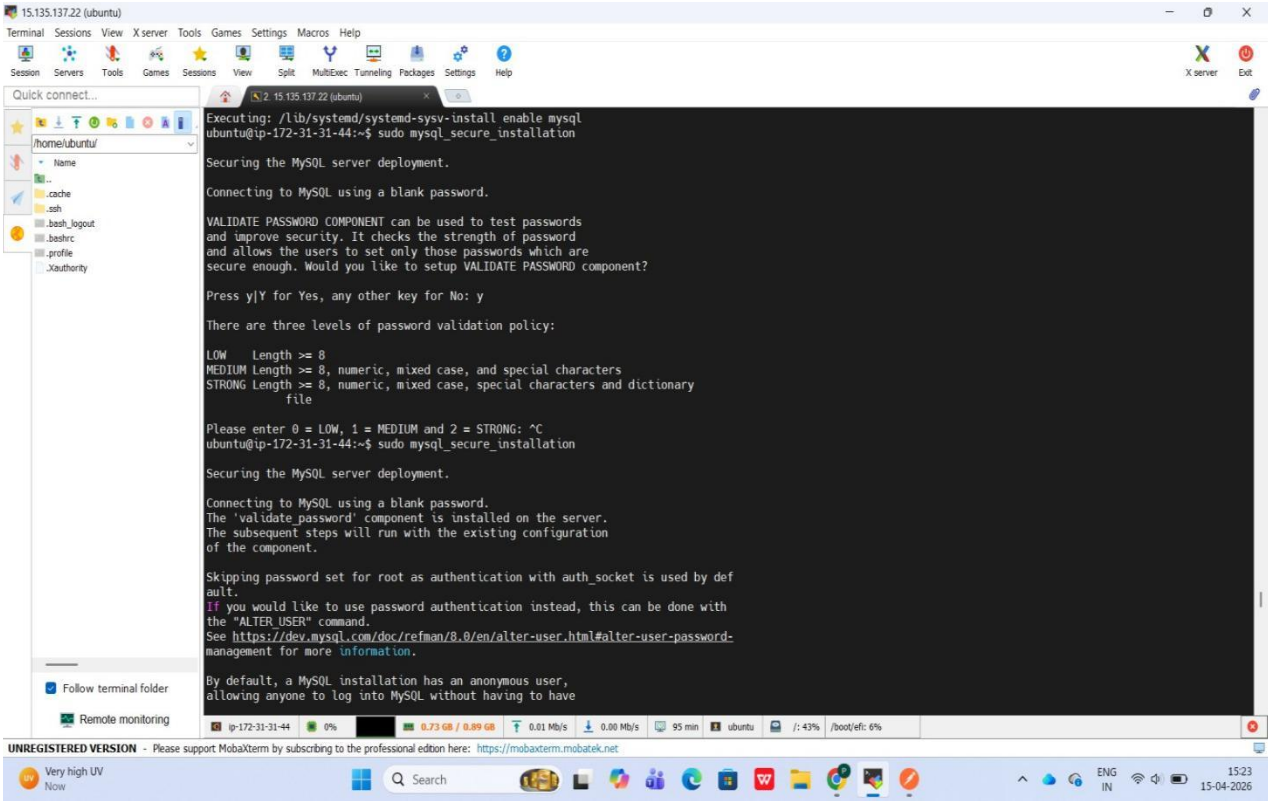
15.135.137.22 (ubuntu)
Executing: /lib/systemd/systemd-sysv-install enable mysql
ubuntu@ip-172-31-31-44:~$ sudo mysql_secure_installation
Securing the MySQL server deployment.
Connecting to MySQL using a blank password.
VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?
Press y|Y for Yes, any other key for No: y
There are three levels of password validation policy:
LOW   Length >= 8
MEDIUM Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary
          file
Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: ^C
ubuntu@ip-172-31-31-44:~$ sudo mysql_secure_installation
Securing the MySQL server deployment.
Connecting to MySQL using a blank password.
The 'validate_password' component is installed on the server.
The subsequent steps will run with the existing configuration
of the component.
Skipping password set for root as authentication with auth_socket is used by def
ault.
If you would like to use password authentication instead, this can be done with
the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-
management for more information.
By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
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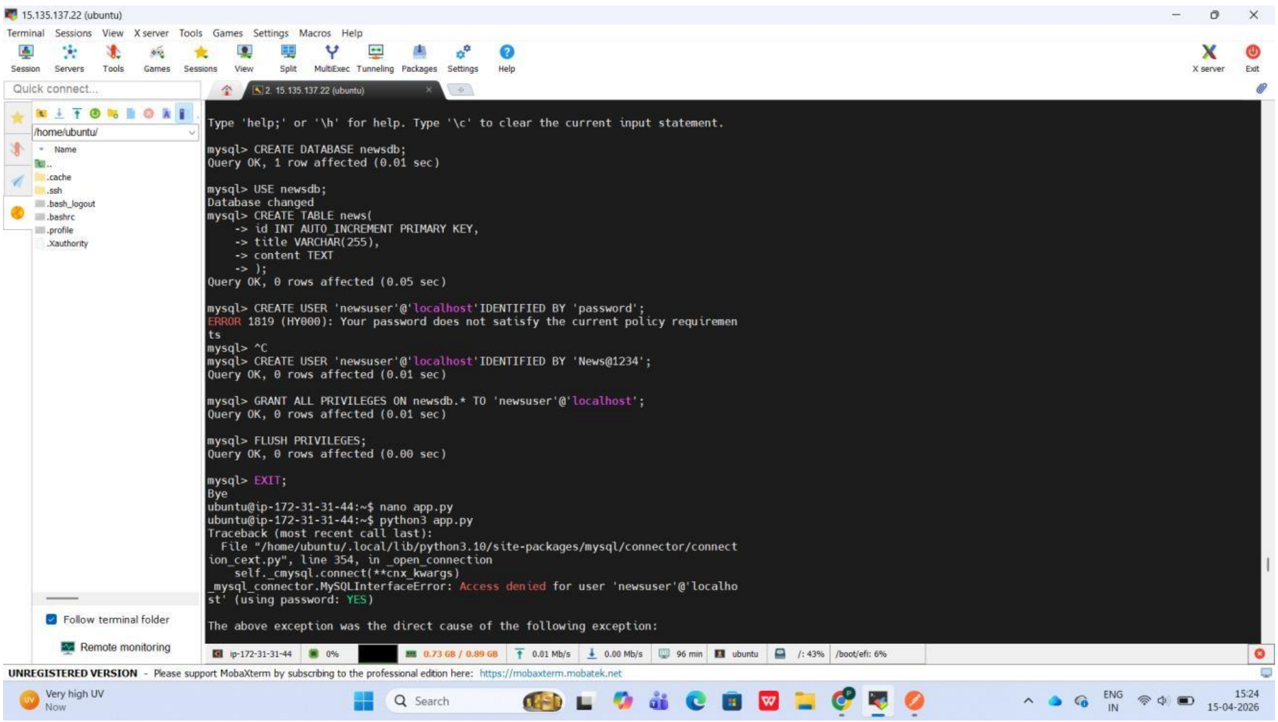
15:23
15-04-2026


15.135.137.22 (ubuntu)
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> CREATE DATABASE newsdb;
Query OK, 1 row affected (0.01 sec)
mysql> USE newsdb;
Database changed
mysql> CREATE TABLE news(
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> title VARCHAR(255),
    -> content TEXT
    ->);
Query OK, 0 rows affected (0.05 sec)
mysql> CREATE USER 'newsuser'@'localhost'IDENTIFIED BY 'password';
ERROR 1819 (HY000): Your password does not satisfy the current policy requiremen
ts
mysql> ^C
mysql> CREATE USER 'newsuser'@'localhost'IDENTIFIED BY 'News@1234';
Query OK, 0 rows affected (0.01 sec)
mysql> GRANT ALL PRIVILEGES ON newsdb.* TO 'newsuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
mysql> EXIT;
Bye
ubuntu@ip-172-31-31-44:~$ nano app.py
ubuntu@ip-172-31-31-44:~$ python3 app.py
Traceback (most recent call last):
  File "/home/ubuntu/.local/lib/python3.10/site-packages/mysql/connector/connect
ion_cext.py", line 354, in _open_connection
    self._cmysql.connect(**cnx_kwargs)
_mysql_connector.MySQLInterfaceError: Access denied for user 'newsuser'@'localho
st' (using password: YES)
The above exception was the direct cause of the following exception:
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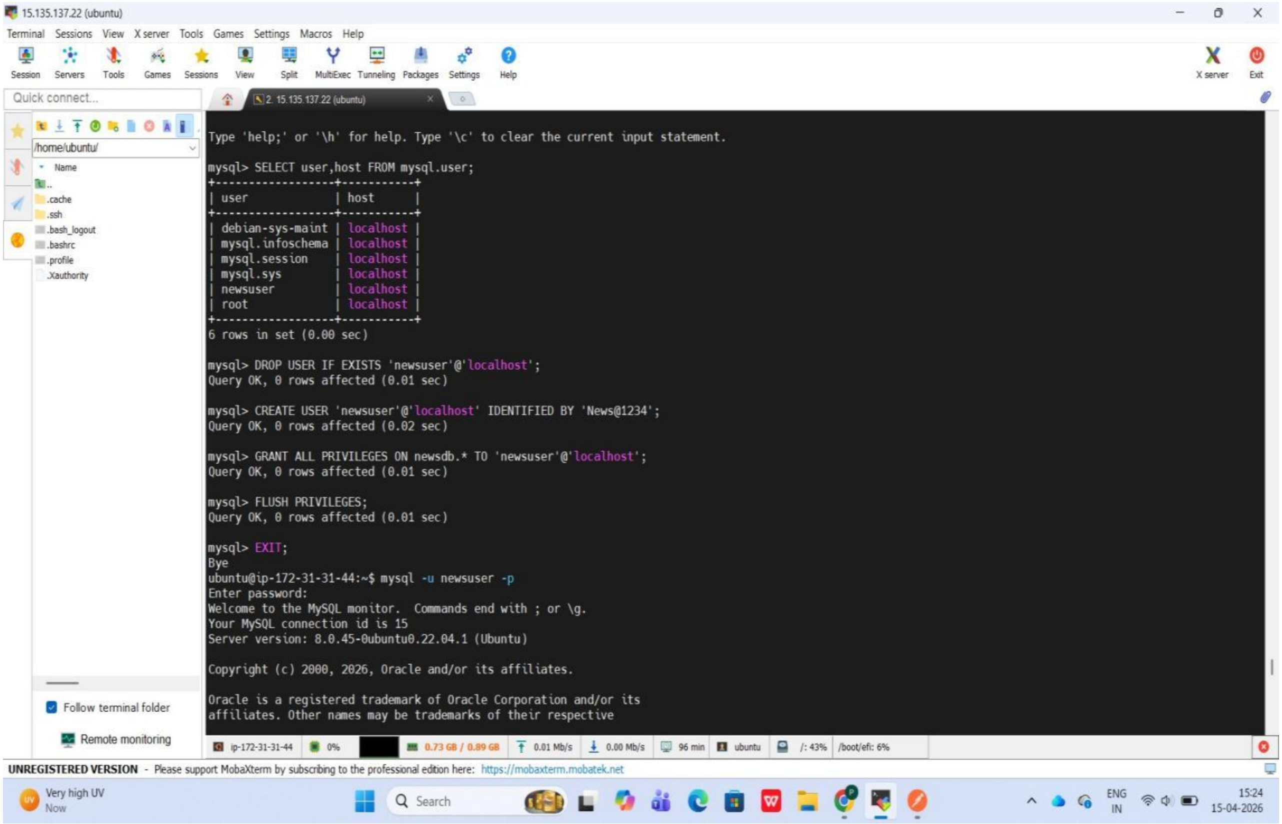
15-04-2026


15.135.137.22 (ubuntu)
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> SELECT user,host FROM mysql.user;
user	host
debian-sys-maint	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
newsuser	localhost
root	localhost
6 rows in set (0.00 sec)
mysql> DROP USER IF EXISTS 'newsuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)
mysql> CREATE USER 'newsuser'@'localhost' IDENTIFIED BY 'News@1234';
Query OK, 0 rows affected (0.02 sec)
mysql> GRANT ALL PRIVILEGES ON newsdb.* TO 'newsuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)
mysql> EXIT;
Bye
ubuntu@ip-172-31-31-44:~$ mysql -u newsuser -p
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)
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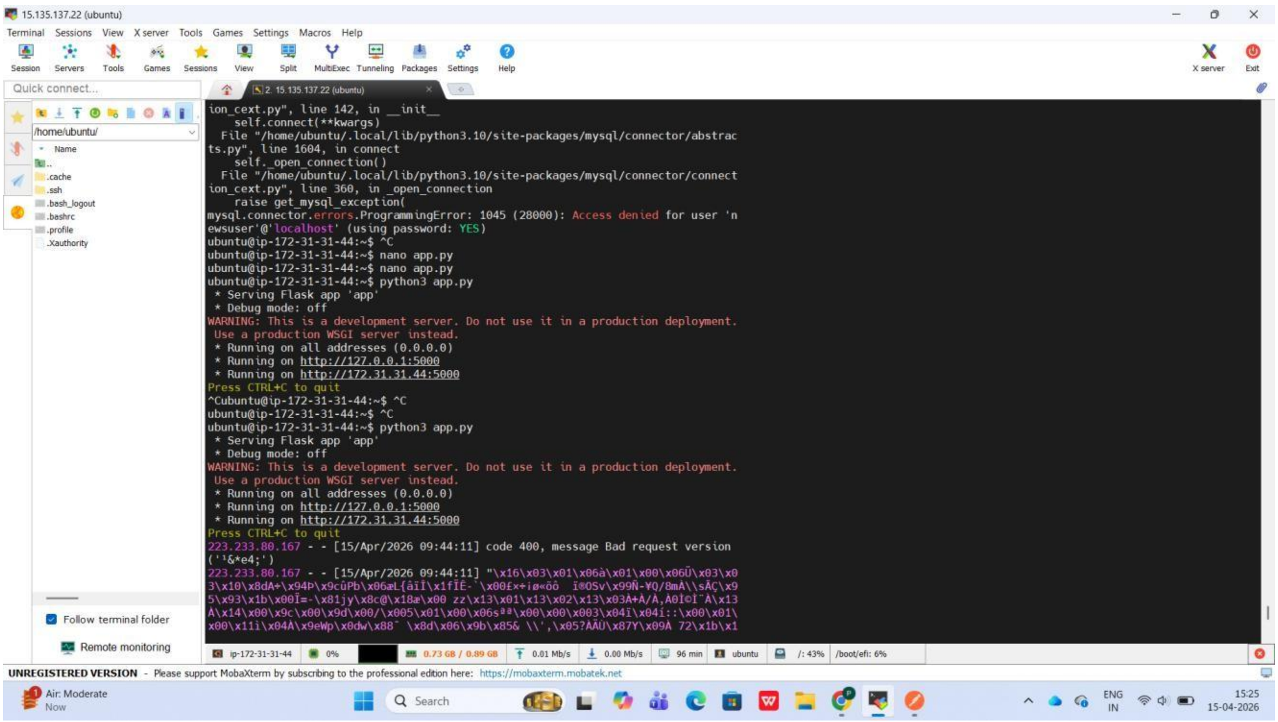
15.135.137.22 (ubuntu)
ion_cext.py", line 142, in __init__
self.connect(**kwargs)
File "/home/ubuntu/.local/lib/python3.10/site-packages/mysql/connector/abstracts.py", line 1604, in connect
self._open_connection()
File "/home/ubuntu/.local/lib/python3.10/site-packages/mysql/connector/connection_cext.py", line 360, in _open_connection
raise get_mysql_exception(
mysql.connector.errors.ProgrammingError: 1045 (28000): Access denied for user 'newsuser'@'localhost' (using password: YES)
ubuntu@ip-172-31-31-44:~$ ^C
ubuntu@ip-172-31-31-44:~$ nano app.py
ubuntu@ip-172-31-31-44:~$ nano app.py
ubuntu@ip-172-31-31-44:~$ python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.31.44:5000
Press CTRL+C to quit
^Cubuntu@ip-172-31-31-44:~$ ^C
ubuntu@ip-172-31-31-44:~$ ^C
ubuntu@ip-172-31-31-44:~$ python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.31.44:5000
Press CTRL+C to quit
223.233.80.167 - - [15/Apr/2026 09:44:11] code 400, message Bad request version
223.233.80.167 - - [15/Apr/2026 09:44:11]
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15:25
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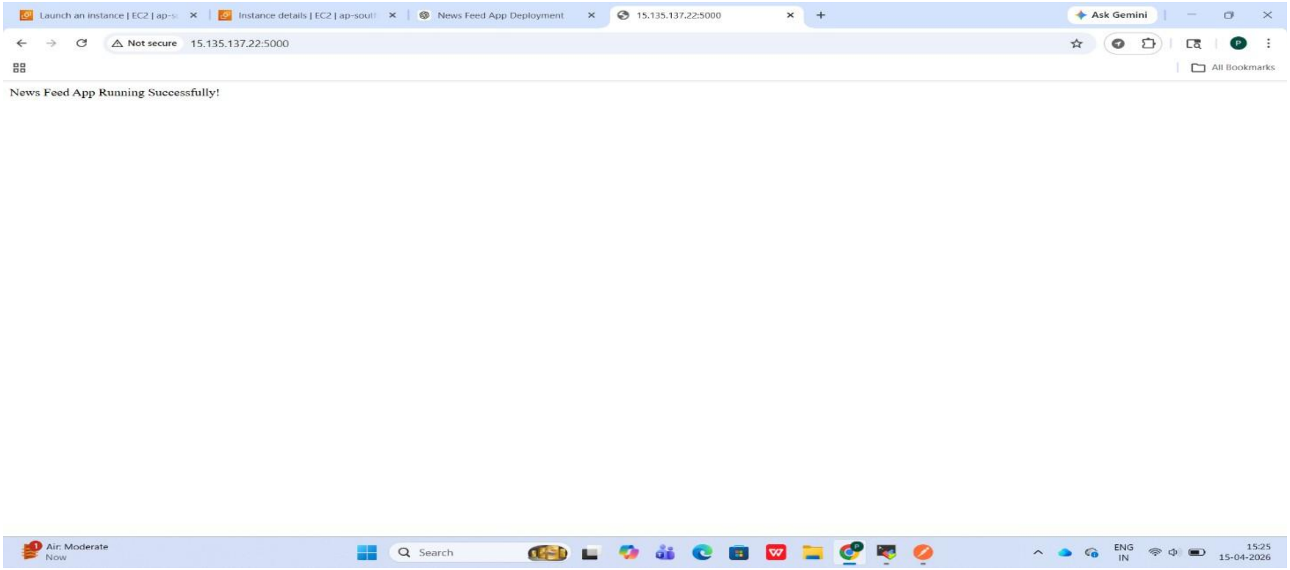
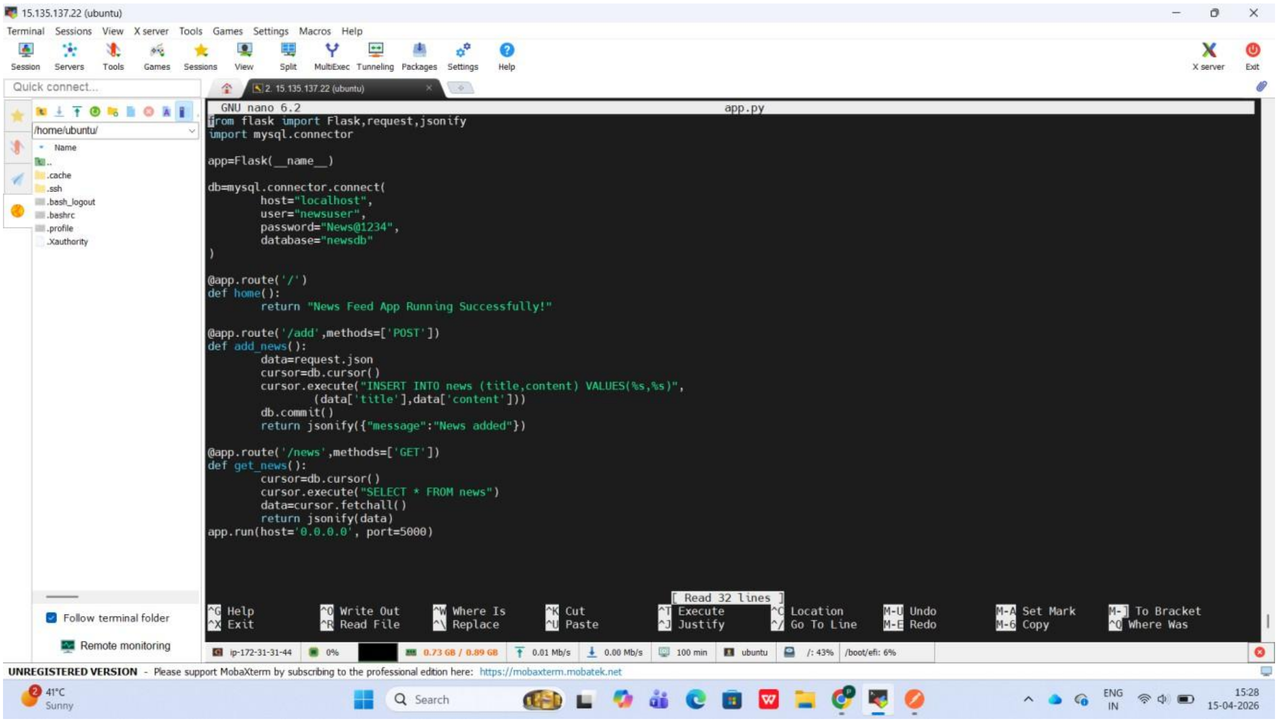

15.135.137.22:5000
Not secure
News Feed App Running Successfully!


15.135.137.22 (ubuntu)
GNU nano 6.2
app.py
from flask import Flask,request,jsonify
import mysql.connector
app=Flask(__name__)
db=mysql.connector.connect(
host="localhost",
user="newsuser",
password="News@1234",
database="newsdb"
)
@app.route('/')
def home():
return "News Feed App Running Successfully!"
@app.route('/add',methods=['POST'])
def add_news():
data=request.json
cursor=db.cursor()
cursor.execute("INSERT INTO news (title,content) VALUES(%s,%s)",
(data['title'],data['content']))
db.commit()
return jsonify({"message":"News added"})
@app.route('/news',methods=['GET'])
def get_news():
cursor=db.cursor()
cursor.execute("SELECT * FROM news")
data=cursor.fetchall()
return jsonify(data)
app.run(host='0.0.0.0', port=5000)
[Read 32 lines]
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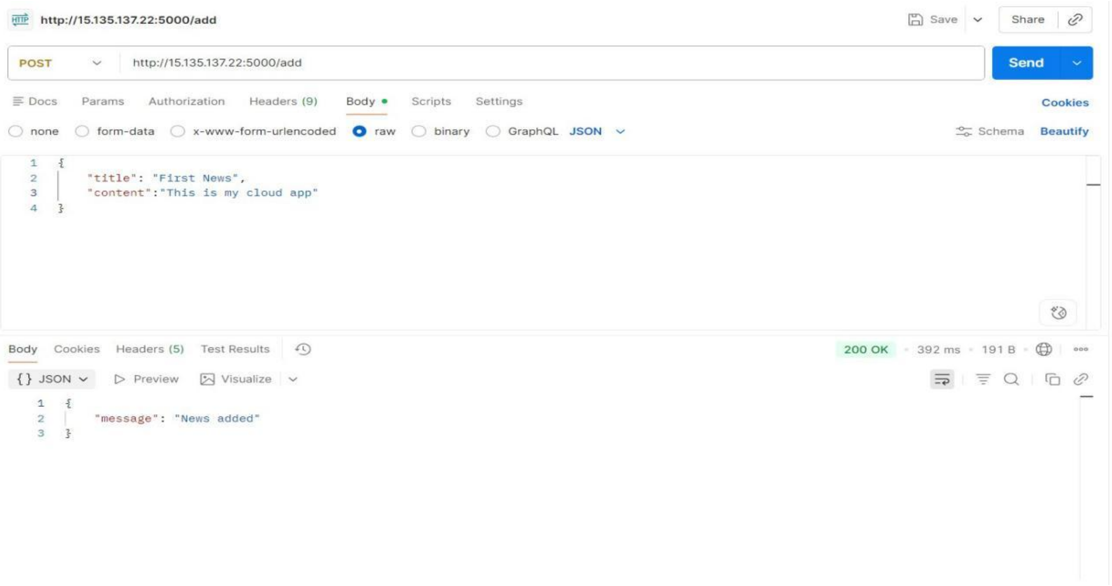

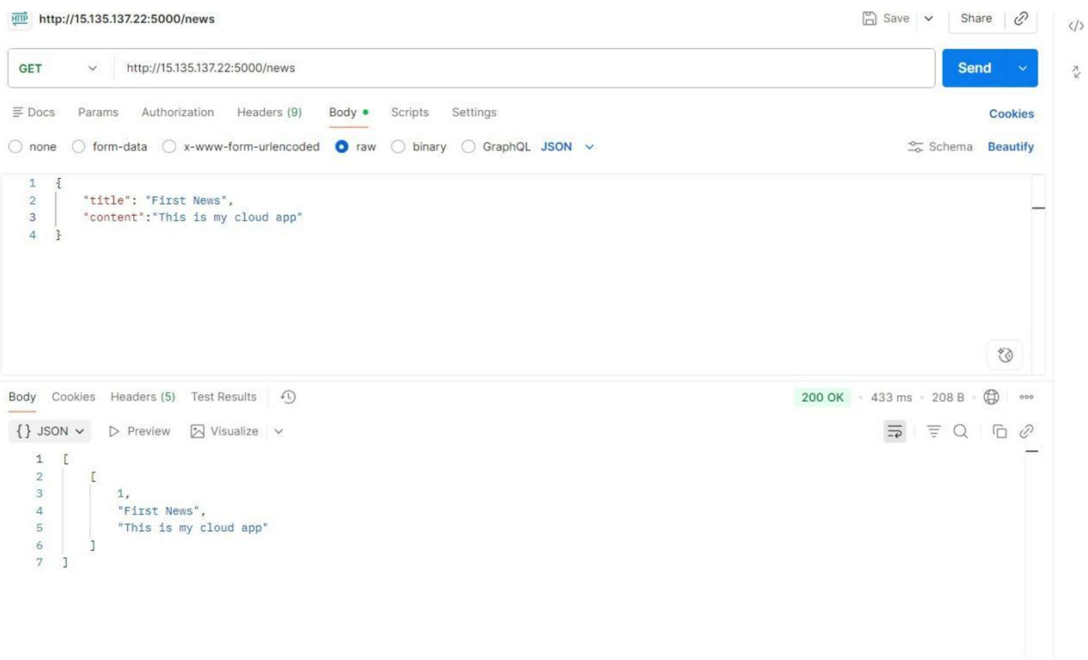

Conclusion:

Deploying a News Feed Application on cloud platforms like Amazon Web Services or Google Cloud Platform enables scalable and efficient data handling. By installing a database on the instance and connecting it with the application, dynamic content can be stored and retrieved easily, demonstrating practical cloud deployment and database integration.

Prof. S. D. Cheke
Course Teacher